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A novel multi-label feature selection method with association rules and rough set

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ABSTRACT

In multi-label learning, each instance is associated with a set of labels. To improve the accuracy and efficiency of multi-label learning tasks, label correlations have been explored. However, existing conventional algorithms obtain label correlation directly from the raw label space, ignoring the significance of the label to the instance. In this study, a novel feature selection method was proposed to select a more relevant and compact feature subset by considering the label distribution and inter-label correlations. First, the concept of label distribution was defined to reflect the significance of each label. Second, a new algorithm for mining association rules was designed to obtain the correlation between labels by improving the existing association rules algorithm. Thus, a new information system was designed by combining the label distribution and correlation between labels. Subsequently, a novel feature selection algorithm was designed in this information system which can effectively remove irrelevant and redundant features in the feature space. Finally, the experimental results demonstrated that the proposed algorithm effectively improves the classification performance and perform better than some state-of-the-art methods.

1. Introduction

In recent studies, owing to the increase in the number of data components, studies using single-label have shown unsatisfactory results in terms of practical application. In some practical problems, including image recognition [1] and protein function prediction [2], a single instance may be simultaneously associated with multiple label variables. Multi-label learning (MLL) is used in cases where a single instance is associated with multiple labels [3]. The aim of MLL is to construct a classifier that can accurately predict the possible labels for the test sets. Multi-label learning approaches can be categorized as follows [4]:

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(1) *Problem transformation methods*, which transform the MLL problem into other learning scenarios. However, this method has several drawbacks. This is a traditional approach that converts multiple label data into single label data. It does not necessarily improve the classification performance because the transformed label contains too many classes, which leads to learning difficulties.
(2) *Algorithm adaptation methods*, which aim to directly solve the MLL tasks using other popular learning algorithms. However, these methods exhibit superior performance only in specific fields, and cannot be applied to solve practical problems on a wider scale.

As mentioned earlier, MLL refers to the construction of a classification model that maps the instances described by feature vectors to multiple class labels. However, the computational cost required for dealing with high-dimensional multi-label data is extremely high. In addition, such data easily lead to a dimension disaster and inhibit the performance of the classification model. Fortunately, the feature dimensionality reduction method is an effective tool for eliminating redundant features, which reduces the computational cost. Therefore, it is highly suitable for MLL.

The association between variables, which is a type of latent information in data, is often used to improve the performance of machine learning models [5]. For example, Liang et al. [6] applied the association information between features to a multi-modal data fusion task, and innovatively proposed an association-based multi-modal fusion method that improves the multi-modal classification performance. Similarly, the generalized performance of multi-label prediction models can be improved using the correlations between the class label variables. Therefore, we can consider using the correlation between labels to improve the performance of classification tasks in MLL. According to the intensity of label correlation mining, MLL can be categorized into three methods: first-order, second-order and high-order [4,5,7]. The first-order strategy ignores the correlations between labels. Generally, this problem is transformed into multiple independent binary classification problems. The second-order strategy is different from the first-order strategy in that it considers the pair associations between labels, such as sorting between related and irrelevant labels. The high-order strategy is different from its first-order and second-order counterparts in many important aspects. It is designed to explore the correlation between one label and other labels. Although the high-order method exhibits the most superior classification performance, owing to its high computing requirements, it has poor scalability. In other words, it cannot be widely used for performing MLL tasks.

Note that although the above methods have the ability to learn from multi-label data, two factors may affect the performance of the algorithm [8,9]. First, the traditional MLL method assumes that all related labels have the same significance for every instance, that is, the effect of each label on the instance is not measured [10]. However, this assumption may not hold true in practical applications. Second, the traditional label correlation is a single strategy. Owing to the limitations of these strategies, the complementary methods that can be utilized for solving the problem of label correlation are scarce.

To address the above problems, a novel way to label an instance x is to specify a real number d_y^x for each possible label y. This would help describe the significance of each label for instance. The description degree of all labels constitutes a data structure similar to that of probability distribution, and is called label distribution [11]. The result of the label distribution depends on the instance properties and labels. During the learning process, we consider the degree to which y describes x. This is called label distribution learning (LDL). LDL solves label ambiguity by using the definition of the label significance description degree. LDL considers labels associated with each instance and attempts to describe the relationship using a model between instances and labels. In other word, LDL uses a set of labels to label an instance, and then learns a mapping function from an instance to all labels [12]. It should be noted that the significance of each label to the instance is not the probability that each label correctly labels the instance, but the proportion of each label that accounts for a full description of the instance. The correlation between labels was obtained from a novel label association rules algorithm, which is an improved form of the classical Apriori algorithm. As a result, label correlation with the mixed-order strategy is obtained.

The rapid expansion of data scale poses huge challenges in terms of storage and mining of data. The large-scale nature of the data is evident from the enormous instance size and numerous dimensions of these data. In addition, this leads to a reduced value density of these data. Selecting valuable information from big data is a top priority. Feature selection or attribute reduct is a core technology for improving the value of big data by reducing their dimensionality [13]. After the feature selection from the original data, the redundant and irrelevant variables in the data are removed, which reduces the storage cost of data and improves their usage value [14]. In classification training, because of the elimination of redundant and irrelevant variables, the training data are more closely related, and in most cases, the prediction performance of the corresponding trained model will be higher than that of the model without feature selection processing [15].

Currently, the mainstream research method is to use feature selection methods to improve the performance of multi-label classifiers while retaining the inherent meaning of features. We propose a frequency distribution multi-label feature selection algorithm with label distributions learning and rough set theory (MLDRS). The most remarkable characteristic of our proposed method is that it does not involve any algorithm transformation learning. It selects a valid feature subset through the label significance distribution and frequent item sets in the label space. First, the significance of each label in the label space is calculated according to the neighborhood granularity for each instance. Second, through an improved version of classical Apriori algorithm, frequent items in the original label space can be further processed to obtain the associated rules that can reflect the labels. In addition, a new information system is designed, an exclusive feature subset is selected for each instance neighborhood that contains only a subset of labels, and the reduct set of features is obtained by merging.

Finally, a multi-label feature selection algorithm is designed, and the experimental results demonstrate that the algorithm proposed in this paper can effectively improve the learning efficiency and classification performance.

According to the methods and approaches discussed earlier, the main contributions of this study are as follows:

(1) It compensates for the disadvantage that the label relationship is not introduced, and eliminates the need to select features of each label individually, which is relatively more efficient to a certain extent.
(2) A new method for mining label association rules is proposed, and the label correlation obtained using this method reflects a mixed order strategy.
(3) Based on the label significant distribution in label space, our proposed method effectively solves high-dimensional problems.

The remainder of this paper is organized as follows: Section 2 briefly reviews several related studies. Section 3 introduces the preparatory knowledge used in the current study. Section 4 introduces the concepts related to the new information system and details the describes the design process of a multi-label feature selection algorithm under this framework. Section 5 details the experimental setup and the results. In Section 6, the conclusions and future work are presented.

2. Related work

2.1. Multi-label learning

The high dimensionality of the feature space and the complexity of label space are the main challenges for MLL. Among these, the high-dimensionality of feature space tends to reduce the classification performance of multi-label classifiers. Feature selection is an important solution that mainly reduces the dimensionality of the feature space based on a certain metric of the original feature set.

To solve multi-label feature selection problems, a method named MLNB was proposed by Zhang et al.[8]. It deals with multi-label instances according to naive Bayes classifiers. It is worth noting that this was the first time that feature selection techniques were introduced into the MLL algorithm. Zhao et al. [16] and Weng et al. [17] proposed a new embedded algorithm by combining label-specific features and global and local correlations based on the idea of LIFT, respectively. These two algorithms compensate for the disadvantage that LIFT has no label correlation, and the learning of global and local correlations further hints at the overall performance of the algorithm. Zhang et al. [18] considered that similar instances share label correlation-based features in the label space. Based on this idea, a new MLL algorithm was proposed by combining label correlation and label-specific features. Che et al. [19] constructed local label correlation and global label correlation based on local attribution reduct with fuzzy rough sets. Che et al.[20] proposed a novel second-order method, by considering the correlation of feature-label, which is represented by the combined kernel, in a new feature space, and then, by using two different strategies to obtain the feature distribution-based label correlation. The relationship between the labels is solved to a certain extent by measuring the relationship between the label variables. Dai et al. [21] proposed two multi-label feature selection methods by learning label symmetric uncertainty and evaluating the feature redundancy.

Mutual information is widely used for feature selection. Li et al. [22] proposed a new granular feature selection method with a maximal correlation minimal redundancy criterion based on mutual information. Hu et al. [23] proposed a feature selection method based on weight selection for the rate of change of undetermined and determined information. Furthermore, based on mutual information and label correlation, Sun et al. [24] proposed a novel feature selection method that constrains convex optimization to obtain an optimal solution. Gonzalez-Lopez et al. [25,26] analyzed mutual information metrics based on a distributed computing environment, and proposed a distributed multi-label feature selection algorithm suitable for the environment. This algorithm enables efficient performance in multi-label classification.

In the learning process, beacuse each instance can have multiple labels, the performance evaluation is highly complex compared to single-label learning. Currently, several popular MLL evaluation metrics have been designed from different perspectives [4]. In our experiments, we chose the Hamming Loss (HL), Ranking Loss (RL), Coverage (CV), Average Precision (AP), One-Error (OE), Macro-F1 (Maf) and Micro-F1 (Mif) as evaluation metrics. These are defined as follows:

(1) Hamming Loss:

$$hloss(h) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{|D|}|h(x_i) \triangle D_i|,$$

where $\triangle$ stands for the symmetric difference between two sets.

(2) Ranking Loss:

$$rloss(f) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{|D_i||\overline{D_i}|}|\{(d', d'')|f(x_i, d') \leqslant f(x_i, d''),\ (d', d'') \in D_i \times \overline{D_i}\}|,$$

where $\overline{D_i}$ denotes the complementary set of D_i in D.

(3) Coverage:

$$coverage(f) = \frac{1}{N}\sum_{i=1}^{N}\max_{d\in D_i} rank_f(x_i, d) - 1.$$

(4) Average Precision:

$$avgprec(f) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{|D_i|}\sum_{d\in D_i}\frac{\left|\left\{d' \in D_i : rank_f\left(x_i, d'\right) \leqslant rank_f(x_i, d)\right\}\right|}{rank_f(x_i, d)}.$$

(5) One-Error

$$one - error(f) = \frac{1}{N}\sum_{i=1}^{N}\left\langle\left[\arg\max_{d\in D} f(x_i, d)\right] \notin D_i\right\rangle.$$

(6) Macro-F1

$$macro_{f1}(h) = \frac{1}{|D|}\sum_{j=1}^{|D|}\frac{2\sum_{i=1}^{N} d_{ij} \cdot h_j(x_i)}{\sum_{i=1}^{N} d_{ij} + \sum_{i=1}^{N} h_j(x_i)}.$$

(7) Micro-F1

$$micro_{f1}(h) = \frac{2\sum_{i=1}^{N}||h(x_i) \cap Y_i||_1}{\sum_{i=1}^{N}||Y_i||_1 + \sum_{i=1}^{N}||h(x_i)||_1},$$

where $||\cdot||$ denotes l_1-norm.

The Hamming loss (HL) is defined by the multi-label classifier $h(\cdot)$, which indicates the number of misclassified instance-label pairs. The others are defined by a real-valued function $f(\cdot)$. Ranking loss (RL) represents the average fraction of incorrectly ordered label pairs; Coverage (CV) indicates the maximum sort distance required to include all category labels in all instances; Average precision is a measure of the average probability of predictive labels, which are actually in the set Y, ordered before a true relevant label $y \in Y$; One-Error (OE) represents the average fraction of the top-rank label that is absent in the relevant label set; Macro-f1 (Maf) indicates the F-measure averaging on each label; and Micro-f1 (Mif) is the global average of F-measures of the prediction results.

Note that for average precision, macro-f1, and micro-f1, the greater the value, the better the performance. For hamming loss, ranking loss, coverage and one-error, the smaller the value, the better the performance.

2.2. Label distribution learning

MLL is an extension of SLL and LDL is an extension of MLL. LDL can learn richer information from data than traditional MLL, which indicates that LDL is a more generalized machine learning framework, in which each instance is labelled by a real-valued vector [27]. Although each label logic value is converted into a real-valued vector, this does not change the nature of the learning process [28,29]. Each instance was still in contact with a set of labels, which did not change the ultimate learning purpose. On the other hand, some differrences exists betweeen label distribution and the original label data. The former is a fundamental value or can be said to be a probability distribution value, while the latter is a logical value. The real value of the label distribution indicates the proportion of one label in all possible labels, whereas the logical value in the original label data indicates whether the label belongs to the instance under consideration [30]. Although this label distribution is not explicitly given, it is generally implicit in instance data. If we can use the appropriate recovery method, it can provide the advantages of label distribution while exploring additional semantic information.

LDL, as introduced earlier in this paper, refers to the process of converting the original logic labels in the instance data into label distributions, depending on the mining of label-related information hidden in instance data. Therefore, different instances typically correspond to different label distributions. The main notations used in this study are as follows.

Let U be a instance space, and Y be a label significance distribution space. For an instance set $X \subseteq U$, the description degree of each label d_j for all instances can be expressed as: $X = \left\{\left(x_1, \left[d_{d_1}^{x_1}, d_{d_2}^{x_1}, \cdots, d_{d_t}^{x_1}\right]\right), \left(x_2, \left[d_{d_1}^{x_2}, d_{d_2}^{x_2}, \cdots, d_{d_t}^{x_2}\right]\right), \cdots, \left(x_s, \left[d_{d_1}^{x_s}, d_{d_2}^{x_s}, \cdots, d_{d_t}^{x_s}\right]\right)\right\}$. Without loss of generality, the description degree of d_j is $d_{d_j}^{x} \in [0, 1]$, satisfying the constrain $\sum_{j=1}^{t} d_{d_j}^{x} = 1$. Set $Y = \{y_1, y_2, \cdots, y_t\}$, where $y_i = \left\{d_{d_i}^{x_1}, d_{d_i}^{x_2}, \cdots, d_{d_i}^{x_s}\right\}$.

3. Preliminary knowledge

To further clarify the content of our discussion, we reviewed several basic concepts in rough set, neighborhood granularity, label distribution and association rules. First, the main notations used in this paper are as follows. U is a non-empty finite set of instances, i.e., $U = \{x_1, x_2, \cdots, x_n\}$, called a universe or instance space. C is a finite non-empty feature space, i.e., $C = \{c_1, c_2, \cdots, c_m\}$ to characterize the instance, and D is a decision feature. If $D = \{d_1, d_2, \cdots, d_t\}$, it means that D has t decision features, which is recorded as D_M. V_a is the value domain of the feature a, and $V = \cup V_a$. And f is an information function of $U \times C \rightarrow V$, where the value of $x \in U$ under the feature a is denoted by $f(x, a)$. The tuple $DIS = (U, C \cup D, V, f)$ represents a decision information system and the tuple $MLDIS = (U, C \cup D_M, V, f)$ represents a multi-label decision information system.

3.1. Rough set

The rough set theory originally proposed by Pawlak [31] is a significant tool for dealing with imprecise and uncertain problems, and has been widely applied in feature selection, pattern recognition, machine learning and other fields. It is a new and rapidly developing tool for solving multi-feature decision problems in both theory and practice. Rough set theory describes the uncertainty of the system through the upper and lower approximation sets, and obtains decision criteria that are easy to handle, robust, and easy to understand through uncertain and imprecise information. The idea of rough set theory comes from the idea of the boundary domain; the uncertain elements are placed in the boundary domain, which is described by the difference between the upper and lower approximation sets. Therefore, the universe can be divided into three domains: positive, negative, and boundary domains. There are two main problems in a rough set: first, how to select features while maintaining the classification ability, and second, how to induce decision classification rules through knowledge reduction.

Definition 1. Let $DIS = (U, C \cup D, V, f)$ be a decision information system. Then, for any $R \subseteq C$, an indiscernible relation $IND(R)$ can be defined as:

$$IND(R) = \{(x, y) \in U \times U | \forall b \in R, f(x, b) = f(y, b)\}. \tag{1}$$

Definition 2. Let $DIS = (U, C \cup D, V, f)$ be a decision information system. The upper and lower approximations of any $X \subseteq U$ are defined as:

$$\overline{R}(X) = \cup\{Y_i \in U/IND(R) | Y_i \cap X \neq \phi\}, \tag{2}$$
$$\underline{R}(X) = \cup\{Y_i \in U/IND(R) | Y_i \subseteq X\}, \tag{3}$$

where $Y_i \in U/IND(R)$ is the partition of indiscernible relation R on U.

The upper and lower approximation sets can also be defined as follows:

$$\overline{R}(X) = \left\{x \in U | [x]_R \cap X \neq \phi\right\}, \tag{4}$$
$$\underline{R}(X) = \left\{x \in U | [x]_R \subseteq X\right\}, \tag{5}$$

where $[x]_R \in U/IND(R)$ represents the equivalence class formed by instance x on indiscernible relation R.

Definition 3. The boundary, positive and negative domains of X with respect to R are respectively defined as follows:

$$BN_R(X) = \overline{R}(X) - \underline{R}(X), \tag{6}$$
$$POS_R(X) = \underline{R}(X), \tag{7}$$
$$NEG_R(X) = U - \overline{R}(X). \tag{8}$$

3.2. Neighborhood granularity

From the view of granular computing, the rough set theory is an effective method for describing data uncertainty. However, a key factor that affects granularity is the size of the granules. Most existing methods depict neighborhood granules by directly specifying their size or using a computational function [32–34]. Because data are often distributed with noisy data and not all data are considered, these methods are suboptimal.

In this section, we present the threshold function used to decrease the uncertainty and show the structure of the data more concretely.

Definition 4. Let $DIS = (U, C \cup D, V, f)$ be a decision information system, for any $B \subseteq C$ and $x_i \in U$. The neighborhood $\delta_B(x_i)$ of x_i in the subset B is defined as:

$$\delta_B(x_i) = \{x_j | x_j \in U, \Delta_B(x_i, x_j) \leqslant \delta\}, \tag{9}$$

where Δ is a metric function. Then, for any $x_1, x_2, x_3 \in U$, the following properties hold,

(1) $\Delta(x_1, x_2) \geqslant 0, \Delta(x_1, x_1) = 0$;
(2) $\Delta(x_1, x_2) = \Delta(x_2, x_1)$;
(3) $\Delta(x_1, x_3) \leqslant \Delta(x_1, x_2) + \Delta(x_2, x_3)$.

$\delta_B(x_i)$ is a neighborhood information granule of x_i. The size of neighborhood is determined by the size of the threshold δ. The larger the δ, the more instances of the neighborhood. Threshold δ can be given externally or calculated according to the instance space U. However, in order to consider as much as possible the actual distribution of the data in feature space, we cite a formula to use the value of instances of the feature space to calculate the threshold δ, which is defined as [30,32,34,35]:

$$\delta = \frac{1}{\omega} Avg\left[\sum_{k=1}^{m} \frac{STDc_k}{\omega}\right], \ \forall c_k \in C, \tag{10}$$

where $STDc_k$ is the standard deviation of the c_k and Avg is the mean of the set. The ω is a tuning parameter whose value can be selected from [0.1,1).

3.3. Label distribution

For the original multi-label data, we cannot obtain the contribution of each label to instance x. Therefore, we first introduce a probability model to describe label significance. In the LDL framework, the label space for each instance is represented by a real-valued vector. Fig. 1 gives a label distribution example for multi-label annotation. For each label y_j, there are $d^x_{y_1} = 0, d^x_{y_2} = 0.3, d^x_{y_3} = 0.2, d^x_{y_4} = 0.4$ and $d^x_{y_5} = 0.1$, they have to satisfy the constraint conditions $d^x_{y_j} \in [0, 1]$ and $\sum_y d^x_{y_j} = 1$. Given a multi-label decision information system $MLDIS = (U, C \cup D_M, V, f)$. Let $d^x_{d_j}$ be the significance description degree of original label d_j to instance x, $[\delta_C(x)]_{d_j}$ denotes the neighborhood of x with the label d_j in the feature space C, $[\delta_C(x)]_{D_M}$ represents the neighborhood of x with the label set D_M in the feature space C. The formula for computing the label significance of each label $d_j \in D_M$ is defined as [10,30]:

$$d^x_{y_j} = \frac{|[\delta_C(x)]_{d_j}|}{|[\delta_C(x)]_{D_M}|}, \tag{11}$$

where label y_j represents the transformation of label d_j in $MLDIS$.

Example 1. Given a multi-label decision information system $MLDIS = (U, C \cup D_M, V, f)$ shown in Table 1. According to Eq. (10) we obtain a neighborhood $\delta_C = 0.208$. Further, we have $\delta_C(x_0) = \delta_C(x_8) = \delta_C(x_9) = \{x_0, x_8, x_9\}, \delta_C(x_4) = \{x_4, x_5, x_6\}$, and the neighborhood of other instances is themselves. Considering instance x_0 with its neighborhood $\delta_C(x_0) = \{x_0, x_8, x_9\}$, where $f(x_0, d_1) = f(x_0, d_2) = f(x_0, d_4) = f(x_8, d_2) = f(x_8, d_3) = f(x_8, d_4) = f(x_9, d_1) = f(x_9, d_2) = f(x_9, d_3) = 1$, consequently $|[\delta_C(x)]_{D_M}| = 9$. Subsequently, we can calculate the significance of each label to x_0. As a result, $d^{x_0}_{d_1} = \frac{2}{9}, d^{x_0}_{d_2} = \frac{3}{9}$ and $d^{x_0}_{d_4} = \frac{2}{9}$. Because of $f(x_0, d_3) = 0, d^{x_0}_{d_3} = 0$. However, $\sum_{j=1}^{4} d^{x_0}_{d_j} = \frac{7}{9} \neq 1$, and then we need futher to process, $d^{x_0}_{d_1} = \frac{2}{9} / \frac{7}{9} = \frac{2}{7}, d^{x_0}_{d_2} = \frac{3}{9} / \frac{7}{9} = \frac{3}{7}$ and $d^{x_0}_{d_4} = \frac{2}{9} / \frac{7}{9} = \frac{2}{7}$. Similarly, the label significance of other instances can also be calculated.Based on the above analysis, the pseudo-code of calculating the label significance can be described by the Algorithm 1.

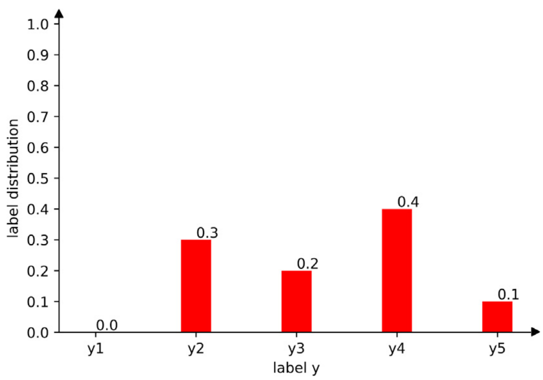

Fig. 1. An example of a multi-label distribution.

Table 1
An example of multi-label distribution.

U	c_1	c_2	c_3	c_4	d_1	d_2	d_3	d_4	y_1	y_2	y_3	y_4
x_0	0.036	0.065	0.082	0.134	1	1	0	1	2/7	3/7	0	2/7
x_1	0.161	0.467	0.096	1.871	0	0	1	1	0	0	1/2	1/2
x_2	0.115	0.336	0.079	0.714	0	1	1	1	0	1/3	1/3	1/3
x_3	0.086	0.142	0.082	1.338	1	1	0	1	1/3	1/3	0	1/3
x_4	0.063	0.140	0.082	0.350	1	1	1	0	1/3	1/6	1/2	0
x_5	0.026	0.041	0.080	0.511	1	0	1	0	1/2	0	1/2	0
x_6	0.102	0.249	0.081	0.499	0	0	1	1	0	0	2/3	1/3
x_7	0.041	0.066	0.083	1.021	0	1	1	1	0	1/3	1/3	1/3
x_8	0.102	0.236	0.080	0.122	0	1	1	1	0	3/7	2/7	2/7
x_9	0.056	0.099	0.085	0.139	1	1	1	0	2/7	3/7	2/7	0
x_{10}	0.038	0.408	0.094	1.437	1	0	1	1	1/3	0	1/3	1/3
x_{11}	0.034	0.063	0.077	1.618	1	1	0	1	1/3	1/3	0	1/3

Algorithm 1. An algorithm for calculating the significance of labels.

Input: A multi-label decision information system $MLDIS = (U, C \cup D_M, V, f)$.
Output: The label significance distribution space Y.
Step 1: Calculate δ and $\delta_C(x)$ for $\forall x \in U$.
Step 2: Calculate $d^x_{d_j}$ for each $x \in U$ by formula:
if $f(x, d_j) = 0$, then $d^x_{d_j} = 0$.
if $\sum_{j=1}^{t} d^x_{d_j} < 1$, then $d^x_{d_j} = \frac{d^x_{d_j}}{\sum_{j=1}^{t} d^x_{d_j}}$.
Step 3: Output Y.
Step 4: End.

Algorithm 1 provides a complete calculation process for the significance of the label. Based on the neighborhood granularity, we first calculate the value of δ and the neighborhood granule of instances (Step 1). Next, we calculate the significance of the labels and determine whether the constraint condition holds true (Step 2). Finally, we obtain the label significance distribution space Y (Step 3). The corresponding time complexity of the algorithm is composed of the following parts: (1) the time complexity of calculating each instance neighborhood is $O(n)$, and (2) that of constructing the label significance distribution space is $O(n \times t)$. Therefore, the total time complexity of Algorithm 1 is $O(n \times (n + t))$.

3.4. Association rules mining

Association analysis is a data analysis technology that is used to mine data associations and correlations in large datasets. The concept of association rules was first proposed by Agrawal, Imielinski, and Swami [36], and is a common technique used for data mining association analysis. In knowledge discovery using large datasets, we can find an association relationship between different projects that we do not know in advance [37,38]. Association rules can be divided according to the dimensionality of data into single dimension and multi-dimension association rules; according to the types of variables processed, they can be divided into Boolean and multivalued feature association rules; and according to the data abstraction level, they can be divided into a single layer and multilayer association rules. Usually, the strength of an association rule is measured by its support and confidence [37]. Support reflects the frequency of related data in the datasets and confidence reflects the probability of another event under the condition of one event, which indicates a conditional probability. Frequent item sets represent a collection of events that often appear together. In MLL, there are used to find the related label feature set and classify the relationship between labels.Mining for association rules are the main technology for data mining, and the Apriori algorithm is a popular search method for frequent item sets in data mining methods. Based on this, many algorithms have been proposed for mining association rules [39–41]. There is a frequent combination of labels in the label space, which are correlations between these labels. These correlations must be fully utilized to improve the classification performance.Based on the aforementioned analysis and referring to the existing algorithms, in this subsection, we propose a new association rules algorithm as an improvement to the classical Apriori algorithm. Using this new algorithm, we further divide the frequent item sets into two parts, namely, frequent related item sets and infrequent unrelated item sets to obtain a new type of associated rules. Labels in frequent related item sets are relevant to each other, whereas those in the infrequent unrelated item sets are irrelevant to each other.In Algorithm 2, we first obtain all frequent item sets using the primitive Apriori algorithm (Steps 1-5). Subsequently, all frequent item sets are divided into the abovementioned two parts. For any two subsets of set F, if the intersection of these two subsets is nonempty, update the value of F_{fre} (Step 6). If the intersection of F_{fre} and any subset of F is empty and the number of elements of the subsets is 1, update the value of F_{unfre} (Step 7). If there exists a label d that does not belong to any subset F, update the value of F_{unfre} (Step 8). Finally, F_{fre} and F_{unfre} are obtained, and a new type of association rules is obtained using Algorithm 2. The time complexity of the algorithm consists of the following parts: (1) In the first part, we obtain all frequent item sets, and the corresponding time complexity is at most $O(t \times k)$; (2) In the second part, we obtain frequent related item sets and infrequent unrelated item sets, and the corresponding time complexity is no more than $O\left(t \times k^2\right)$. Therefore, the maximum total time complexity for Algorithm 2 is $O\left(t \times \left(k + k^2\right)\right)$.From the above algorithm, we obtain the mixed-order label correlations. As shown in Example 2, L_1 and L_2 are second-order label correlations, and L_3 is first-order label correlations. If the number of labels is increased or the parameter is decreased, then higher order correlations can be obtained. Therefore, the improved new algorithm includes first-order, second-order and high-order correlations simultaneously, which further enriches the relevance exploration method of labels.

Algorithm 2. An algorithm for mining all association rules.

Input: A multi-label decision information system $MLDIS = (U, C \cup D_M, V, f)$ and parameter *min_support*.
Output: Label association set F_{fre} and F_{infre}.
Step 1: Scanning the data and generate candidate *1*-itemsets C_1.
Step 2: According to the *min_sup*, generate frequent *1*-item sets F_1.
Step 3: Cycle from *2*-item sets, and generate frequent *k*-item sets from frequent *k-1*-item sets.
Step 4: Connecting and pruning operation was executed by F_k to generate a candidate *k+1*-item sets C_{k+1}.
Step 5: Repeat step 3,4 until $F_{k+1} = \phi$. Then, all frequent item sets $F = F_1 \cup F_2 \cup \cdots \cup F_k$.
Step 6: Let $F_{fre} = \phi, F_{infre} = \phi$.
  While $F \neq \phi$:
  For each $L_i, L_j \in F$:
    If $L_i \cap L_j \neq \phi$:
      $F_{fre} = F_{fre} \cup \left(L_i \cup L_j\right)$
      $F = F - L_i - L_j$
Step 7: For each $L_i \in F$:
  If $L_i \cap F_{fre} = \phi$ and $|L_i| = 1$:
    $F_{infre} = F_{infre} \cup L_i$
    $F = F - L_i$
Step 8: For each $d \in D$:
  If all $L_i \in F, d \notin L_i$:
    $F_{infre} = F_{infre} \cup \{d\}$
Step 9: Output F_{fre} and F_{infre}.
Step 10: End.

In rough set theory, the information system is an important knowledge expression system. The concepts of knowledge core and knowledge reduction are based on the information system. We develop a new information system that is defined as follows:

Definition 5. A multi-label decision significance distribution system is defined as $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$ where,

(1) U is a finite non-empty set of instances;
(2) C is a finite non-empty feature space;
(3) Y is a label significance distribution space;
(4) $V = \cup V_a, V_a$ is a set of values for feature a;
(5) f is an information function, $f : U \times A \rightarrow V$ such that for all $x \in U$ and $a \in A, f(x, a) \in V_a$;
(6) L is the associated set of the label space, i.e., $L = F_{fre} \cup F_{infre}$.

The other concepts of the multi-label decision significance distribution system will be introduced in Section 4.

4. Feature selection based on a new rough set model

The label distribution, which contains a certain number of labels, indicates the extent to which each label describes an instance [11]. The original label space can only reflect whether one instance has this label, and no other information can be known. In the above section, we used LDL to convert the original label space into the label significance distribution space, which not only reflects what labels an instance has, but also obtains the significance of these labels for that instance. However, we characterize the label correlation based on association rules from two aspects: frequent related item sets and infrequent unrelated item sets, and simultaneously obtain the label correlation that includes first-order, second-order, and higher-order correlations. It can be found that the research object of these two works is label space, and the purpose is to use more label space information to improve the classification performance of subsequent feature selection. We have now obtained label decision significance distribution sets and label association rules. To solve the impact of high dimensional data on the classifier, we used the framework of label distribution learning and related rules to represent instance information in the data. In this section, we define other concepts based on the multi-label decision significance distribution system, and use the label space correlation rules to design a feature selection algorithm.

4.1. A new rough set model on MLDS-DS

From the above process, we have defined a multi-label decision significance distribution system. Then, we extend some concepts and properties of classical rough set theory to the multi-label decision significance distribution system, and design a new rough set theory model. Based on the new theoretical framework, we study the perforamce of multi-label feature selection and classification.

Definition 6. Let $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$ be a multi-label decision significance distribution system, and $d^x_{d_j}$ is a significance degree of label d_j to instance x. Then, $[X]^\beta_{L_i}$ is defined as:

$$[X]^\beta_{L_i} = \left\{x | \forall d_j \in L_i, d^x_{d_j} \geqslant \beta, \quad L_i \subseteq L\right\}, \tag{12}$$

where $[X]^\beta_{L_i}$ denotes the instance x contained in the subset L_i of label distribution space, and the label significance is greater than β.

Definition 7. Let $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$ be a multi-label decision significance distribution system, for any $[X]^\beta_{L_i} \subseteq U$ and $B \subseteq C$. The upper and lower approximations of $[X]^\beta_{L_i}$ are defined as:

$$\overline{apr}_B\left([X]^\beta_{L_i}\right) = \left\{x | \forall x \in U, \delta_B(x) \cap [X]^\beta_{L_i} \neq \phi\right\}, \tag{13}$$

$$\underline{apr}_B\left([X]^\beta_{L_i}\right) = \left\{x | \forall x \in U, \delta_B(x) \subseteq [X]^\beta_{L_i}\right\}. \tag{14}$$

The approximation set have the following properties:

(1) $\underline{apr}_B\left([X]^\beta_{L_i}\right) \subseteq [X]^\beta_{L_i} \subseteq \overline{apr}_B\left([X]^\beta_{L_i}\right)$;

(2) $\underline{apr}_B(\phi) = \overline{apr}_B(\phi) = \phi$, $\underline{apr}_B(U) = \overline{apr}_B(U) = U$;

(3) $\overline{apr}_B\left([X]^\beta_{L_i} \cup [Y]^\beta_{L_i}\right) = \overline{apr}_B\left([X]^\beta_{L_i}\right) \cup \overline{apr}_B\left([Y]^\beta_{L_i}\right)$;

(4) $\underline{apr}_B\left([X]^\beta_{L_i} \cap [Y]^\beta_{L_i}\right) = \underline{apr}_B\left([X]^\beta_{L_i}\right) \cap \underline{apr}_B\left([Y]^\beta_{L_i}\right)$;

(5) $[X]^\beta_{L_i} \subseteq [Y]^\beta_{L_i} \Rightarrow \overline{apr}_B\left([X]^\beta_{L_i}\right) \subseteq \overline{apr}_B\left([Y]^\beta_{L_i}\right)$, $\underline{apr}_B\left([X]^\beta_{L_i}\right) \subseteq \underline{apr}_B\left([Y]^\beta_{L_i}\right)$;

(6) $\overline{apr}_B\left([X]^\beta_{L_i} \cap [Y]^\beta_{L_i}\right) \subseteq \overline{apr}_B\left([X]^\beta_{L_i}\right) \cap \overline{apr}_B\left([Y]^\beta_{L_i}\right)$;

(7) $\underline{apr}_B\left([X]^\beta_{L_i} \cup [Y]^\beta_{L_i}\right) \supseteq \underline{apr}_B\left([X]^\beta_{L_i}\right) \cup \underline{apr}_B\left([Y]^\beta_{L_i}\right)$;

(8) $\overline{apr}_B\left(\sim [X]^\beta_{L_i}\right) = \sim \underline{apr}_B\left([X]^\beta_{L_i}\right)$;

(9) $\underline{apr}_B\left(\sim [X]^\beta_{L_i}\right) = \sim \overline{apr}_B\left(\sim [X]^\beta_{L_i}\right)$.

Then, the positive domain of $[X]^\beta_{L_i}$ with respect to B is defined as:

$$POS_B\left([X]^\beta_{L_i}\right) = \left\{x | \forall x \in U, \delta_B(x) \subseteq [X]^\beta_{L_i}\right\}. \tag{15}$$

Definition 8. Let $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$ be a multi-label decision significance distribution system, $B \subseteq C$, the dependency of L_i to B is defined as:

$$\gamma_B(L_i) = \frac{|POS_B\left([X]^\beta_{L_i}\right)|}{|U|}, \tag{16}$$

$\gamma_B(L_i)$ reflects the degree of L_i depends on B. Clearly, $0 \leqslant \gamma_B(L_i) \leqslant 1$, the larger the positive domain, the stronger the ability of condition attribution set B to describe L_i. If $\gamma_B(L_i) = 1$, L_i completely depends on B.

Theorem 1. Given a multi-label decision significance distribution system $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$. If $\beta_1 \leqslant \beta_2$, then $POS_B\left([X]_{L_i}^{\beta_2}\right) \subseteq POS_B\left([X]_{L_i}^{\beta_1}\right)$.

Proof. If $\beta_1 \leqslant \beta_2$, we have $[X]_{L_i}^{\beta_2} \subseteq [X]_{L_i}^{\beta_1}$. $\left(\delta_B(x) \cap [X]_{L_i}^{\beta_2}\right) \subseteq \left(\delta_B(x) \cap [X]_{L_i}^{\beta_1}\right)$ when $[X]_{L_i}^{\beta_2} \subseteq [X]_{L_i}^{\beta_1}$, which can get $\overline{apr}_B\left([X]_{L_i}^{\beta_2}\right) \subseteq \overline{apr}_B\left([X]_{L_i}^{\beta_1}\right)$, and $\underline{apr}_B\left([X]_{L_i}^{\beta_2}\right) \subseteq \underline{apr}_B\left([X]_{L_i}^{\beta_1}\right)$ can be proved. So, $POS_B\left([X]_{L_i}^{\beta_2}\right) \subseteq POS_B\left([X]_{L_i}^{\beta_1}\right)$ is obvious.

Lemma 1. Let $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$ be a multi-label decision significance distribution system. If $\beta_1 \leqslant \beta_2 \leqslant \cdots \leqslant \beta_m$, then $POS_B\left([X]_{L_i}^{\beta_m}\right) \subseteq \cdots \subseteq POS_B\left([X]_{L_i}^{\beta_2}\right) \subseteq POS_B\left([X]_{L_i}^{\beta_1}\right)$.

Theorem 2. Given a multi-label decision distribution table $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$ and parameter β, if $B_1 \subseteq B_2 \subseteq C$, then we have:

(1) $POS_{B_1}\left([X]_{L_i}^{\beta}\right) \subseteq POS_{B_2}\left([X]_{L_i}^{\beta}\right)$.
(2) $\gamma_{B_1}(L_i) \leqslant \gamma_{B_2}(L_i)$.

Proof. If $B_1 \subseteq B_2$, obviously, we have $\delta_{B_2}(x) \subseteq \delta_{B_1}(x)$. Assume that $\delta_{B_1}(x) \subseteq \underline{apr}_{B_1}\left([X]_{L_i}^{\beta}\right)$, then we have $\delta_{B_2}(x) \subseteq \underline{apr}_{B_2}\left([X]_{L_i}^{\beta}\right)$. There may be x, $\delta_{B_1}(x) \not\subseteq \underline{apr}_{B_1}\left([X]_{L_i}^{\beta}\right)$ and $\delta_{B_2}(x) \subseteq \underline{apr}_{B_2}\left([X]_{L_i}^{\beta}\right)$. Therefore, $POS_{B_1}\left([X]_{L_i}^{\beta}\right) \subseteq POS_{B_2}\left([X]_{L_i}^{\beta}\right)$ and $\gamma_{B_1}(L_i) \leqslant \gamma_{B_2}(L_i)$.

Definition 9. Let $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$ be a multi-label decision significance distribution system, $B \subseteq C$, we say the feature subset B is a reduct if:

(1) $\gamma_B(L_i) = \gamma_C(L_i)$,
(2) $\forall c \in B, \gamma_B(L_i) > \gamma_{B-c}(L_i)$.

The first condition ensures that $POS_B\left([X]_{L_i}^{\beta}\right) = POS_C\left([X]_{L_i}^{\beta}\right)$. The second condition ensures that there is no redundant feature in the reduct. Therefore, this definition guarantees a reduct is the minimal subset of features that has the same characterizing power as the whole feature set.

Definition 10. Let $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$ be a multi-label decision significance distribution system, $B \subseteq C$ and $\forall c \in C - B$, the significance of feature c with respect to B can be defined as:

$$Sig(c, B, L_i) = \gamma_{B\cup c}(L_i) - \gamma_B(L_i). \tag{17}$$

That indicates the necessity of the feature c with respect to the feature subset B. The feature c is redundant in B with respect to L_i, if $Sig(c, B, L_i) = 0$. Otherwise, the feature c is necessary for B.

Example 2. For the Table 1, let the value of $min\_support = 0.5$, the results of frequent item sets are obtained by Algorithm 2, i.e., $F = \{\{d_1\}, \{d_2\}, \{d_3\}, \{d_4\}, \{d_2, d_4\}, \{d_3, d_4\}\}$, $F_{fre} = \{\{d_2, d_4\}, \{d_3, d_4\}\}$ and $F_{infre} = \{\{d_1\}\}$. Then $L = \{\{d_2, d_4\}, \{d_3, d_4\}, \{d_1\}\}$. For convenience, let $L_1 = \{d_2, d_4\}$, $L_2 = \{d_3, d_4\}$ and $L_3 = \{d_1\}$. From Definition 6 and set $\beta = 0.3$, we can get $[X]_{L_1}^{\beta} = \{x_2, x_3, x_7, x_{10}\}$, $[X]_{L_2}^{\beta} = \{x_1, x_6, x_7, x_{10}\}$ and $[X]_{L_3}^{\beta} = \{x_3, x_4, x_5, x_{10}, x_{11}\}$. Further, according to Definition 7, each approximate result is obtained, i.e., $\overline{apr}_C\left([X]_{L_1}^{\beta}\right) = \{x_2, x_3, x_7, x_{11}\}$, $\underline{apr}_C\left([X]_{L_1}^{\beta}\right) = \{x_2, x_3, x_7, x_{11}\}$, $\overline{apr}_C\left([X]_{L_2}^{\beta}\right) = \{x_1, x_6, x_7, x_{10}\}$, $\underline{apr}_C\left([X]_{L_2}^{\beta}\right) = \{x_1, x_6, x_7, x_{10}\}$, $\overline{apr}_C\left([X]_{L_3}^{\beta}\right) = \{x_3, x_4, x_5, x_6, x_{10}, x_{11}\}$ and $\underline{apr}_C\left([X]_{L_3}^{\beta}\right) = \{x_3, x_{10}, x_{11}\}$. This example clearly shows how to get the upper and lower approximations for each target set.

4.2. A novel feature selection algorithm for MLDS-DS

In this subsection, we propose a frequent distribution multi-label feature selection algorithm with label distribution learning and rough set theory (MLDRS). This algorithm includes label significance, label correlation, and feature dependency. The aim of the proposed algorithm is to use inter-label correlation to select unique features for each label association subset. In other words, the algorithm performs feature selection by considering the relationship between labels based on the label significance distribution. Following these ideas, the details of the design procedure of the multi-label feature selection algorithm are formulated in Algorithm 3.

Algorithm 3. An algorithm (MLDRS) for finding a reduct.

Input: A multi-label decision significance distribution system $MLDS\text{-}DS = (U, C \cup Y, V, f, L)$.
Output: A reduct Red.
Step 1: Initialization $Red = \phi$.
Step 2: For each $L_i \in L$:
  $Red_{L_i} = \phi$
  If $\delta_C(x) \in [X]_{L_i}$:
    $POS_C\left([X]_{L_i}^{\beta}\right) = POS_C\left([X]_{L_i}^{\beta}\right) \cup \{x\}$
Step 3: Compute the feature dependency $\gamma_C(L_i)$.
Step 4: For $\forall c_i \in C - Red_{L_i}$, if $\delta_{Red_{L_i} \cup c_i}(x) \subseteq [X]_{L_i}^{\beta}$:
  $POS_{Red_{L_i} \cup c_i}\left([X]_{L_i}^{\beta}\right) = POS_{Red_{L_i} \cup c_i}\left([X]_{L_i}^{\beta}\right) \cup \{x\}$
Step 5: Compute the feature dependence $\gamma_{Red_{L_i} \cup c_i}(L_i)$.
Step 6: Compute $Sig(c_i, Red_{L_i}, L_i)$.
Step 7: If $Sig(c_i, Red_{L_i}, L_i) > 0$:
  $Red_{L_i} = Red_{L_i} \cup c_i$
Step 8: $Red = Red \cup Red_{L_i}$.
Step 9: Return Red.
Step 10: End.

In Algorithm 3, we obtain a feature subset, which means that the elements in this feature subset can represent the description of the instance by all features in the multi-label decision significance distribution system. The time complexity is analyzed as follows: Steps 2-3, we find the positive domain and calculate the feature dependence of each label association subset, and its time complexity is $O(tn)$. In Steps 4-9, we explore the best features until Step 7 does not hold ture, and its time complexity is $O(m^2)$. Consequently, the overall time complexity is $O(tnm^2)$.

5. Experiments

In this section, we conduct a series of experiments to verify the effectiveness and robustness of the proposed algorithm and compare it with several other MLL algorithms based on different metrics. Experiments are executed under the environment of Windows 10, Intel Core i5-8250U CPU and 8GB DDR4 and all algorithms in our experiments are implement in Spyder 5.1.5 on Anaconda 2.1.4.

5.1. Datasets and experimental settings

To illustrate the performance of our methods and the comparative methods, we selected nine multi-label datasets. All the datasets were obtained from the Mulan Library[1] and the description of all datasets is displayed in Table 2. We conducted experiments to evaluate the performance of MLDRS, and the experimental results were compared with those of other existing multi label learning algorithms, i.e. ML-KNN [42], RF-ML [43],PPTReliefF [43], MLNB [8], RReliefF [43], MDFS [44], JFSC [45] and MVLD [46], and the details are as follows:

- ML-KNN: A multi-label classification method, which is used for evaluating the classification performance of all multi-label algorithms. The parameter $k = 10$ is the number of nearest neighbours. The global time complexity of the algorithm is $O(n^2m + tnk + nm + tk)$.
- MLNB: A feature selection algorithm based on principal component analysis and genetic algorithm. The parameter value of PCA is set to the same value as the number of features selected by MLDRS.

[1] http://mulan.sourceforge.net/datasets.html

Table 2
Characteristics of multi-label datasets.

Data sets	Instances	Features	Labels	Density	Domain
Emotions	593	72	6	0.311	Music
Flags	194	19	7	0.485	Image
CAL500	502	68	174	0.150	Music
Birds	645	260	19	0.053	Audio
Scene	2407	294	6	0.179	Image
Yeast	2417	103	14	0.303	Biology
Image	2000	294	5	0.247	Image
Education	5000	550	33	0.044	Text
Computer	5000	681	33	0.046	Text

Table 3
The size of selected features and the reduct rate for nine datasets by MLDRS algorithm.

Data sets	Original features	Selected features	Rate(%)
Emotions	72	26	36.11%
Flags	19	8	42.11%
CAL500	68	18	26.47%
Birds	260	77	29.62%
Scene	294	84	28.57%
Yeast	103	60	58.25%
Image	294	135	45.92%
Education	550	307	55.82%
Computer	681	296	43.47%

- RF-ML, RReliefF and PPTReliefF: These three methods are based on the promotion of ReliefF algorithm. We adjust the parameters so that the number of features selected by these algorithms is the same as the number of features selected by MLDRS. The time complexity of these three algorithms does not exceed $O(n^2m)$.
- MDFS: It is used to find distinctive features among multiple class labels. The parameter α is set to $1, \beta$ and γ are searched in $\left\{10^{-3}, 10^{-2}, \cdots, 10^{3}\right\}$, and k is fixed to $q-1$.
- JFSC: It proposed an algorithm for learning features and constructing classifiers through tag correlation. The parameters α, β and γ are searched in $\left\{4^{-5}, 4^{-4}, \cdots, 4^{5}\right\}, \eta$ is searched in $\{0.1, 1, 10\}$, and $\tau = 0.5$. The global time complexity of the algorithm is $O(nm^2 + m^2t + mt^2 + nmt)$.
- MVLD: It is a view-based approach to construct a view-specific label learning model and a classifier by maximizing label feature dependency. The parameter λ_1, λ_2 are searched in $\left\{10^{-7}, 10^{-6}, \cdots, 10^{-1}\right\}$, the search ranges of parameters α_3, α_4 and $\sigma$are $\left\{10^{-5}, 10^{-4}, \cdots, 10^{1}\right\}, \left\{10^{3}, 10^{4}, \cdots, 10^{6}\right\}$ and $\{0.5, 1.0, 5.0, 10\}$, respectively. The global time complexity of the algorithm is $O(q(n^3 + n^2m) + n^3)$, where q is the number of iterations.
- MLDRS: A frequent division multi-label (MLDRS) feature selection method with label distributions learning and rough set theory. Let $\omega = 0.9$, and β is set to the range $[0.15, 0.25]$. The maximum global time complexity of the algorithm is $O\left(n^2 + t\left(n + k + k^2 + nm^2\right)\right)$.

Ten-fold cross-validation was conducted on all datasets. The original data were randomly divided into ten parts, each with a similar number of instances. In each fold, a part is used to test, and the remainder is used for training learning algorithms. We iterated ten times to ensure that each part can be used as a test set, and the average measurement of the results of ten runs is considered as the final result of the algorithm. We selected the HL, RL, CV, AP, OE, Maf and Mif to indicate the classification performance. These seven evaluation metrics indicate the performance of MLL systems from different perspectives.

5.2. Experimental results

To illustrate the effectiveness of MLDRS, we compared it with ML-KNN, RF-ML, PPTReliefF, MLNB, RReliefF, MDFS, JFSC and MVLD with respect to predictive classification performance. The experimental methods can be divided into two categories: the necessity of feature selection and the effectiveness of the algorithm. To systematically verify the effectiveness of MLDRS, we performed comparative analyses in two experimental scenarios: the same number of features and the state-of-the-art MLL methods.Tables 4–10 show the predictive classification performance for different evaluation metrics. In these tables, the underlining represents the global best predictive classification performance of the comparison algorithm,

Table 4
Performance comparison with nine algorithms in Hamming Loss (↓).

	Set 1	Set 2				Set 3			Our algorithm
Data sets	ML-KNN	RF-ML	PPTReliefF	MLNB	RReliefF	MDFS	JFSC	MVLD	MLDRS
Emotions	0.2684×0.0153	0.2861×0.0260	0.2807×0.0214	0.2664×0.0215	0.2553×0.0223	0.2286×0.0164	0.2137×0.0176	0.2437×0.0057	0.2105×0.0177
Flags	0.3263×0.0275	0.3229×0.0483	0.3374×0.0476	0.3308×0.0581	**0.2897**×**0.0406**	0.6640×0.0122	0.2771×0.0419	0.2725×0.0335	0.3137×0.0270
CAL500	0.1543×0.0040	0.1381×0.0064	0.2076×0.0073	0.2521×0.0193	0.1372×0.0059	0.1413×0.0007	0.1376×0.0009	0.1387×0.0027	0.1364×0.0060
Birds	0.0575×0.0091	0.0533×0.0082	0.0637×0.0072	0.0842×0.0154	0.0533×0.0082	0.0541×0.0018	0.0508×0.0030	0.0540×0.0071	**0.0514**×**0.0050**
Scene	0.0864×0.0036	0.1819×0.0095	0.2097×0.0351	0.1941×0.0371	0.2995×0.0398	**0.0942**×**0.0143**	0.1105×0.0044	0.8209×0.0013	**0.1253**×**0.0257**
Yeast	0.2249×0.0099	0.2894×0.0089	0.2958×0.0069	0.3045×0.0112	0.3047×0.0128	0.2077×0.0064	0.2018×0.0014	0.1878×0.0054	0.1802×0.0094
Image	0.1744×0.0046	0.2724×0.0226	0.3319×0.0436	**0.2096**×**0.0093**	0.4705×0.0614	0.6658×0.3736	**0.2095**×**0.0035**	0.7527×0.0042	0.2105×0.0514
Education	0.0385×0.0006	0.0454×0.0006	0.0612×0.0013	0.0513×0.0006	0.1585×0.0045	0.0957×0.0009	0.0382×0.0008	0.0380×0.0010	0.0327×0.0005
Computer	0.0397×0.0018	0.0477×0.0014	0.0534×0.0012	0.0505×0.0029	0.2380×0.0059	0.9870×0.0021	0.0354×0.0010	0.0342×0.0016	0.0309×0.0013
Avg.Rank	4.67	5.28	7.33	6.67	6.06	6.11	2.44	4.33	2.11

Table 5
Performance comparison with nine algorithms in Ranking Loss (↓).

	Set 1	Set 2				Set 3			Our algorithm
Data sets	ML-KNN	RF-ML	PPTReliefF	MLNB	RReliefF	MDFS	JFSC	MVLD	MLDRS
Emotions	0.1995×0.0226	0.6386×0.0661	0.5311×0.0412	0.4556×0.0371	0.6218×0.0328	0.1810×0.0186	0.2048×0.0358	0.2405×0.0151	0.1807×0.0224
Flags	0.2456×0.0396	0.5090×0.0658	0.5547×0.0774	0.5043×0.0816	0.4692×0.0560	0.2385×0.0151	0.2234×0.0411	0.2109×0.0277	0.2087×0.0534
CAL500	0.7330×0.0150	0.7871×0.0140	0.6964×0.0127	0.6838×0.0141	0.7747×0.0148	0.1900×0.0016	0.1801×0.0018	0.1796×0.0042	0.1762×0.0046
Birds	0.5118×0.0666	0.5441×0.0642	0.3577×0.0576	0.4102×0.0567	0.5441×0.0642	0.1258×0.0107	0.2169×0.0209	0.1043×0.0169	**0.2242×0.0238**
Scene	0.6587×0.1652	0.9861×0.0191	0.6178×0.0922	0.4810×0.0872	0.4865×0.0901	0.0909×0.0239	0.0966×0.0049	0.0813×0.0066	**0.1478×0.0320**
Yeast	0.5036×0.0248	0.5680×0.0161	0.5760×0.0114	0.5585×0.0175	0.5419×0.0191	0.1829×0.0088	0.1740, 0.0039	0.1494×0.0057	0.1485×0.0088
Image	0.1860×0.0072	0.9460×0.0200	0.7121×0.0978	**0.2217×0.0123**	0.6648×0.0636	0.1862×0.1105	0.2139×0.0095	0.1503×0.0120	0.2294×0.0646
Education	0.0778×0.0020	0.9834×0.0046	0.7056×0.0065	**0.0882×0.0029**	0.4828×0.0145	0.0911×0.0025	0.0878×0.0051	0.0705×0.0048	0.0955×0.0049
Computer	0.0866×0.0035	0.7674×0.0180	0.5766×0.0149	0.0570×0.0029	0.4896×0.0098	0.0895×0.0026	0.0969×0.0048	**0.0658×0.0038**	0.0879×0.0042
Avg.Rank	4.89	8.61	7.44	5.11	6.94	3.44	3.56	1.89	3.11

Table 6
Performance comparison with nine algorithms in Coverage (↓).

	Set 1	Set 2				Set 3			Our algorithm
Data sets	ML-KNN	RF-ML	PPTReliefF	MLNB	RReliefF	MDFS	JFSC	MVLD	MLDRS
Emotionst	1.7705×0.2308	5.4547×0.2822	4.6001×0.2656	4.1196×0.2830	5.0836×0.3000	1.9623×0.1056	1.3397×0.0421	1.3592×0.0148	**1.9259**×**0.1047**
Flags	6.1071×0.2417	6.2534×0.2031	6.2514×0.2823	6.6871×0.1772	6.1311×0.2156	4.0987×0.1287	4.5400×0.0282	4.5449×0.0243	3.9124×0.3026
CAL500	130.6×1.917	130.5×1.211	130.9×1.405	130.86×1.403	130.5×1.823	130.2984×0.7032	130.7566×0.0099	130.7574×0.0117	129.9917×3.2695
Birds	3.0908×1.2793	10.3376×1.2202	8.3294×1.2690	8.8721×1.1707	10.3376×1.2202	3.4291×0.2258	3.1504×0.0248	3.1341×0.0259	2.8217×0.5962
Scene	0.4958×0.8088	5.9356×0.0904	4.1897×0.4558	3.4871×0.4415	**0.3515**×**0.4561**	0.5559×0.1207	0.0952×0.0048	0.0827×0.0061	0.8206×0.1699
Yeast	6.6999×0.3075	12.6408×0.1906	12.6850×0.1691	12.1932×0.1521	11.8946×0.2030	6.5439×0.1426	6.4597×0.0036	6.4309×0.0101	6.4580×0.1811
Image	1.0175×0.0274	4.9300×0.0762	4.0344×0.3706	**1.1680**×**0.0621**	3.8334×0.1972	2.8842×1.2568	1.2252×0.0093	**1.1746**×**0.0117**	1.1845×0.3258
Education	3.4562×0.0910	32.5906×0.1057	25.3918×0.2793	6.0328×1.0584	18.8305×0.4277	3.9485×0.0843	3.1252×0.0061	3.0999×0.0057	**4.0724**×**0.2512**
Computer	0.6734×0.0708	0.9220×0.4756	0.6410×0.4662	0.4480×0.4610	0.8470×0.4728	0.3816×0.1180	0.1444×0.0047	0.1016×0.0062	**0.1918**×**0.2176**
Avg.Rank	3.89	8.00	7.40	6.30	6.56	4.11	3.11	2.56	3.00

Table 7
Performance comparison with nine algorithms in Average Precision (↑).

	Set 1	Set 2				Set 3			Our algorithm
Data sets	ML-KNN	RF-ML	PPTReliefF	MLNB	RReliefF	MDFS	JFSC	MVLD	MLDRS
Emotions	0.7578×0.0125	0.4596×0.0454	0.5327×0.0315	0.5654×0.0306	0.5448×0.0207	0.7793×0.0170	0.7646×0.0323	0.7173×0.0168	0.7891×0.0329
Flags	0.6423×0.0308	0.6575×0.0533	0.5789×0.0510	0.5692×0.0588	0.6762×0.0429	0.8153×0.0193	0.8013×0.0451	0.8089×0.0322	0.8221×0.0389
CAL500	0.2549×0.0064	0.2594×0.0104	0.2211×0.0098	0.2126×0.0094	0.2559×0.0101	0.4793×0.0028	0.5107×0.0084	0.5091×0.0038	**0.4874×0.0114**
Birds	0.5348×0.0600	0.5093×0.0576	**0.6221×0.0534**	0.5460×0.0560	0.5093×0.0576	0.6973×0.0181	0.5554×0.0226	0.3097×0.0214	0.5092×0.0291
Scene	0.8636×0.0076	0.1904×0.0174	0.4242×0.0730	0.4045×0.0560	0.4000×0.0570	0.8490×0.0354	0.8364×0.0080	0.8655×0.0082	**0.7669×0.0464**
Yeast	0.7622×0.0178	0.4794×0.0103	0.4723×0.0121	0.4941×0.0147	0.4985×0.0161	0.7439×0.0106	0.7591×0.0070	0.7840×0.0063	0.7883×0.0092
Image	0.7771×0.0090	0.2651×0.0147	0.4061×0.0596	**0.7367×0.0136**	0.3609×0.0488	0.5910×0.0627	0.7362×0.0143	0.8140×0.0134	0.7301×0.0711
Education	0.6117×0.0084	0.0600×0.0043	0.2342×0.0024	0.2562×0.0725	0.2178×0.0116	0.5478×0.0123	0.6171×0.0115	0.6462×0.0167	0.6489×0.0106
Computer	0.6435×0.0077	0.2687×0.0172	0.3556×0.0092	0.3671×0.0637	0.2149×0.0064	0.6352×0.0215	0.6795×0.0063	0.7132×0.0085	**0.6433×0.0125**
Avg.Rank	4.11	7.72	6.88	6.33	7.06	3.67	3.11	2.89	3.06

Table 8
Performance comparison with nine algorithms in One Error (↓).

	Set 1	Set 2				Set 3			Our algorithm
Data sets	ML-KNN	RF-ML	PPTReliefF	MLNB	RReliefF	MDFS	JFSC	MVLD	MLDRS
Emotions	0.3036×0.0324	0.6089×0.04918	0.4832×0.0637	**0.3267×0.0080**	0.4652×0.0575	0.3161×0.0261	**0.3154×0.0499**	0.4133×0.0405	0.3320×0.0729
Flags	0.2751×0.0693	0.2111×0.0514	0.2892×0.0960	0.2910×0.6777	0.2729×0.0600	0.2796×0.0465	0.2661×0.0973	0.2223×0.0710	**0.2157×0.0615**
CAL500	0.1255×0.0585	0.3527×0.0548	0.4842×0.0769	0.1774×0.6470	0.4822×0.0788	0.1359×0.0171	0.1176×0.0196	0.1037×0.0244	0.1025×0.0148
Birds	0.4776×0.0641	0.9752×0.0058	0.7938×0.0330	**0.3994×0.3120**	0.8434×0.0301	0.3741×0.0231	0.4918×0.0415	0.7256×0.0223	0.5681×0.0406
Scene	0.2356×0.0204	0.8039×0.1632	0.6187×0.1501	**0.2801×0.1324**	0.6370×0.2330	0.2479×0.0558	0.2696×0.0146	0.2202×0.0143	0.3781×0.0827
Yeast	0.2346×0.0227	0.4650×0.0314	0.5155×0.0238	0.3159×0.0125	0.4663×0.0438	0.2511×0.0086	0.2218×0.0183	0.2081×0.0068	0.2025×0.0134
Image	0.3410×0.0144	0.8350×0.2316	0.6940×0.0675	**0.4055×0.0253**	0.7500×0.2026	0.5692×0.0531	0.4160×0.0280	0.2915×0.0226	0.4155×0.1115
Education	0.5028×0.0191	0.8624×0.0066	0.7205×0.0127	**0.5804×0.0079**	0.7332×0.0099	0.5876×0.0203	0.4948×0.0185	0.4646×0.0196	0.6192×0.0139
Computer	0.4322×0.0157	0.4778×0.0120	0.4798×0.0153	0.2930×0.0144	0.4993×0.0087	0.4513×0.0323	0.3852×0.0119	0.3568×0.0106	0.2386×0.0147
Avg.Rank	3.33	7.44	7.78	4.56	7.67	4.44	3.56	2.78	3.44

Table 9
Performance comparison with nine algorithms in Macro-F1 (↑).

	Set 1	Set 2				Set 3			Our algorithm
Data sets	ML-KNN	RF-ML	PPTReliefF	MLNB	RReliefF	MDFS	JFSC	MVLD	MLDRS
Emotions	0.6123×0.0647	0.1505×0.2147	0.5505×0.1498	0.5592×0.1426	0.6608×0.1698	**0.5881×0.0545**	0.5499×0.0414	0.4833×0.0212	0.5641×0.0625
Flags	0.5343×0.0274	0.4912×0.3900	0.5234×0.2692	0.6657×0.0245	**0.6830×0.3266**	0.8770×0.0081	0.4717×0.0524	0.5341×0.0332	0.4971×0.0207
CAL500	0.1013×0.0109	0.0533×0.1964	0.1753×0.2259	0.0788×0.0035	0.2072×0.2761	0.0611×0.0042	0.0837×0.0281	0.0647×0.0104	**0.0869×0.0081**
Birds	0.2036×0.0697	0.0008×0.0082	0.2609×0.2965	0.2732×0.1212	0.5439×0.4019	0.1501×0.0287	0.1233×0.0569	0.1503×0.0260	**0.0874×0.0208**
Scene	0.7308×0.0207	0.0282×0.0607	0.2520×0.2786	**0.6504×0.0462**	0.3823×0.3674	0.7021×0.0831	0.5920×0.0204	**0.7397×0.0177**	0.3522×0.0526
Yeast	0.3572×0.0131	0.1901×0.3424	0.3807×0.2231	0.3049×0.0121	0.4676×0.1866	0.3236×0.0332	0.3375×0.0026	**0.4312×0.0099**	0.3217×0.0094
Image	0.5589×0.0170	0.0542×0.0872	0.4325×0.2211	0.5305×0.0164	0.6713×0.2555	0.2655×0.0442	0.3496×0.0124	**0.6357×0.0152**	0.3989×0.1294
Education	0.2447×0.0426	0.0027×0.0102	0.1114×0.1462	**0.3109×0.0592**	0.2678×0.2875	0.9779×0.0005	0.1993×0.0451	0.1605×0.0052	0.1720×0.0483
Computer	0.2117×0.0304	0.1708×0.0939	0.1738×0.1883	0.3095×0.0149	**0.3922×0.2520**	0.9817×0.0010	0.1619×0.0362	0.1556×0.0127	0.1642×0.0329
Avg.Rank	3.33	8.56	5.11	4.11	2.00	4.11	6.44	5.11	5.44

Table 10
Performance comparison with nine algorithms in Micro-F1 (↑).

	Set 1	Set 2				Set 3			Our algorithm
Data sets	ML-KNN	RF-ML	PPTReliefF	MLNB	RReliefF	MDFS	JFSC	MVLD	MLDRS
Emotions	0.6478×0.0191	0.1583×0.1873	0.5381×0.1477	0.5882×0.2134	0.5940×0.1297	0.6195×0.0380	0.6075×0.0375	0.5508×0.0094	0.6663×0.0462
Flags	0.6829×0.0107	0.4564×0.3411	0.5168×0.2590	**0.7556**×**0.1502**	0.5413×0.2605	0.8790×0.0076	0.6899×0.0510	0.7192×0.0418	0.6789×0.0199
CAL500	0.3179×0.0086	0.0497×0.1696	0.1588×0.2012	0.3183×0.0143	0.1700×0.2175	0.3267×0.0089	0.2630×0.0054	0.3669×0.0253	0.3740×0.0117
Birds	0.2784×0.0218	0.0016×0.0151	0.2382×0.2649	**0.4803**×**0.0294**	0.1589×0.1478	0.5038×0.0222	0.2126×0.0672	0.3905×0.0322	0.1719×0.0288
Scene	0.7279×0.0193	0.0364×0.0850	0.2164×0.2563	**0.6491**×**0.0654**	0.2491×0.3169	0.7057×0.0766	0.6035×0.0200	0.7305×0.0165	0.5882×0.1011
Yeast	0.6372×0.0175	0.1827×0.2931	0.3759×0.2246	0.5991×02234	0.4193×0.2148	0.6170×0.0209	0.6366×0.0026	0.6802×0.0095	**0.6144**×**0.0105**
Image	0.5605×0.0215	0.0491×0.0768	0.2712×0.2050	**0.5328**×**0.0162**	0.3188×0.2827	0.3668×0.0695	0.3913×0.0181	**0.6344**×**0.0208**	0.4823×0.1234
Education	0.3264×0.0187	0.0048×0.0176	0.1136×0.1469	**0.3672**×**0.0112**	0.1026×0.1488	0.4793×0.0005	0.3458×0.0170	0.4779×0.0127	0.1249×0.0140
Computer	0.4142×0.0156	0.1720×0.0961	0.1781×0.1889	**0.5161**×**0.0151**	0.1015×0.1493	0.5828×0.0009	0.4816×0.0093	0.5363×0.0128	0.1041×0.0194
Avg.Rank	3.56	8.78	7.33	3.67	7.22	2.56	4.56	2.44	4.89

and the black and blue highlights represent the local best performance in the two different experimental scenarios. For each evaluation metric, the symbols "↓" and "uparrow" indicate "the smaller, the better" and "the larger, the better", respectively.

5.2.1. Necessity of feature selection

Feature selection is an important data preprocessing technology that not only reduces the cost of data storage but also improves the usage value of data. However, in practical applications, the necessity of this technology also needs to be demonstrated. In this subsection, we analye the MLL classification results with and without feature selection through experiments.ML-KNN is a multi-label classification method that is used as a classifier to classify and predict the original data directly (Set 1). Across all the 63 configurations (i.e., nine data sets $\times$ seven evaluation metrics), ML-KNN ranks 1^{st} among the nine compared algorithms for only 6.3% cases, 2^{nd} for 18.8% cases, and 3^{rd} for 12.7% cases. Although ML-KNN has the best classification performance in the first case, it is not significantly different from the other feature algorithms. For example, the difference between ML-KNN and the optimal feature algorithm MDFS was only 0.0078 in the HL metric of the Scene dataset. On the other hand, in the statistical test, other feature selection algorithms had significant advantages over ML-KNN. The observation results show that the classification performance of the dataset was improved in most cases after feature selection. Therefore, it is necessary to use a feature selection method to improve the classification performance.

5.2.2. Experiment results with the same number of features

In this subsection, we verify the advantages of MLDRS by setting the compared algorithms to the same number of features selected by MLDRS (Set 2). Table 3 shows the final size of selected features of MLDRS algorithm in each dataset and the proportion of selected features in each dataset. Except the Yeast and Education datasets, the proportion of selected features in other data sets does not exceed 50%.In this subsection, we choose RF-ML, PPTReliefF, MLNB and RReliefF algorithms to perform experimental comparison with MLDRS, and set the same number of selected features, as shown in Table 3. The purpose of this is to test whether the feature set obtained by the MLDRS algorithm can improve the classification performance of MLL tasks. Among the HL, RL, CV and AP metrics (Tables 4–7), the performance of the above four comparison algorithms ranks 1^{st} in 22.2 % cases on the all datasets, whereas MLDRS ranks 1^{st} place in 77.8 % cases.However, for the OE, Maf and Mif metrics (Tables 8–10), the above four comparison algorithms performed significantly better than MLDRS. MLDRS ranks in 1^{st} place among the five algorithms in only 18.5% cases. MLNB algorithm ranks 1^{st} in OE and Mif metrics, in 55.6 % and 66.7 % cases, respectively. In particular, in the Maf metric, RReliefF ranks 1^{st} in 77.8% cases on all data sets. This shows that MLDRS prefers to judge the performance of algorithms in terms of HL, RL, CV and AP metrics.Overall, among all 63 configurations, MLDRS ranks 1^{st} among all metrics at 49.2%. Furthermore, in the statistical test, the four comparison algorithms rank relatively low, thus indicating that MLDRS is more advantageous than them.Through the experimental analysis discussed in this subsection, under the same number of features, the proposed algorithm showed approximately half of the overall advantages compared with the other four algorithms, which demonstrates the effectiveness of the algorithm.

5.2.3. Experiment results with the state-of-the-art MLL methods

In this subsection, we verify the advantages of MLDRS by comparing it with several recent MLL methods (Set 3). We chose the MDFS, JFSC and MVLD algorithms as comparative experiments for the MLDRS. For these three algorithms considered for comparison, we used their default parameter settings for the experiments. The purpose of this study is to test whether MLDRS performs better than the recently proposed algorithm.Overall, among 63 configurations, the best performance of MLDRS accounted for 39.7%. Although the proportion of MLDRS optimality was not large, the difference between MLDRS and the optimal algorithm was not significant. For example, for the Birds and Image datasets, MLDRS is a suboptimal algorithm.

5.2.4. Execution time

During the experiment, we recorded the execution time of each algorithm on different datasets. Fig. 2 shows the execution time of all algorithms on each dataset. Except the CAL500 dataset, MLDRS runs the longest, while on other datasets, MLDRS runs in the middle range. We observed that ML-KNN generally has the lowest execution time, especially for larger datasets. RF-ML,RRTRelifF, and RReliefF are all generalizations of ReliefF algorithm, and they have the same time complexity, so they have similar execution time. MLNB has the longest execution time in most small datasets, whereas MDFS has the longest execution time in big datasets.

5.2.5. Generality

In this subsection, we make a simple summary of the experimental results. In experiments, we study the performance of MLDRS on two scenarios of experimental methods. In the above four subsections, the experimental results were analyzed and summarized from different perspective and different experimental settings, and the execution time of all algorithms was recorded. The proposed algorithm combines label distribution learning and rough set theory. First, LDL transforms the logical label space into real-value label space, which reflects the significance degree of the labels to the instances. Second, we obtain the target set $[X]_{L_i}^{\beta}$ with rich label information through association rules L and parameter β. Finally, based on a novel rough set model, we use more reliable label information and instance information to select features and get a more compact features subset.Observing the experimental results in these tables (Tables 4–10), we find that in the second exper-

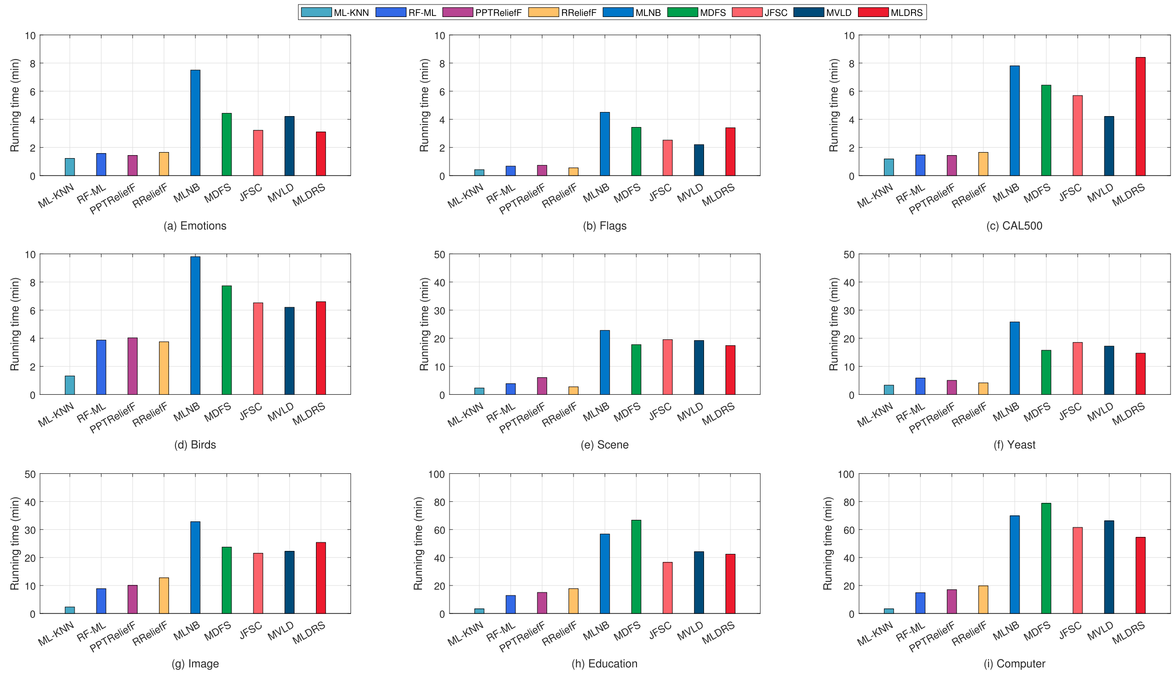

Fig. 2. Execution time on all datasets.

imental scenario, the global optimal predictive classification performance of the comparison algorithm is more widely distributed in the HL, RL, CV and AP metrics. While for the OE, Maf and Mif metrics, the global optimal performance is mainly distributed in the first experimental scenario. In general, compared with other algorithms, the performance of MLDRS algorithm on all datasets has been improved to a certain extent. The above experimental results show that the classification performance of MLDRS algorithm is better than other algorithms to some extent under different evaluation metrics and different datasets.

5.3. Statistical test

To statistically compare the performance of nine algorithms on nine datasets, the two step statistical hypothesis test method proposed by Demiar [47] is used. First, the *Friedman test* is applied to the original hypothesis that "the average ranking of all algorithms is the same". On this basis, in the second step of the method, the *Nemenyi test* is used to test the average ranking compared to each other. The results can be represented by CD (critical difference) diagram.

$$F_F = \frac{(N-1)\chi_F^2}{N(k-1)-\chi_F^2}, \text{ where,}$$

$$\chi_F^2 = \frac{12N}{k(k+1)}\left(\sum_{j=1}^{k} R_j^2 - \frac{k(k+1)^2}{4}\right),$$

Given k comparison algorithms and N multi-label datasets. Let r_i^j be the rank of the j-th algorithms on the i-th datasets. Let $R_j = \frac{1}{N}\sum_{i=1}^{N} r_i^j$ be the average rank for j-th algorithm on all algorithms. The Friedman statistical is distributed according to χ_F^2 with $k-1$ degrees of freedom, which is distributed according to the F-distribution with $k-1$ and $(k-1)(N-1)$ degrees of freedom. The average rank of all methods on different evaluation metrics is displayed in Table . As shown in Table 11, the

Table 11
Friedman statistics on different evaluation metrics.

Evaluation metrics	F_F	Critical value ($\alpha = 0.05$)
Hamming Loss	6.1657	2.0868
Ranking Loss	16.5209	
Coverage	10.2742	
Average Precision	20.3403	
One-Error	10.1424	
Macro-F1	4.1329	
Micro-F1	17.6700	

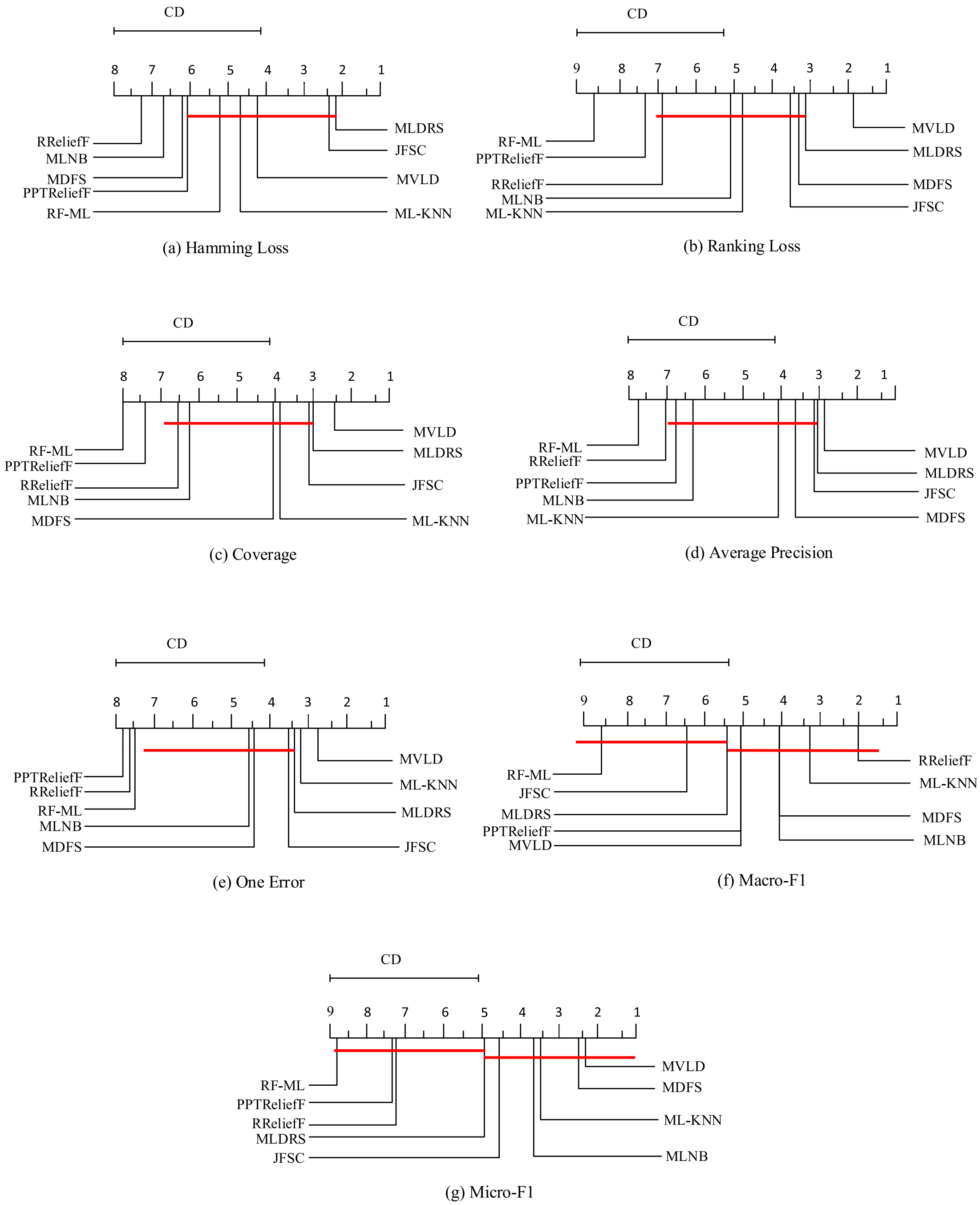

Fig. 3. The results of Nemenyi test.

value of F_F on different evaluation measures is illustrated. Moreover, with the critical value at significance level $\alpha = 0.05$, we have $CV_\alpha = 2.0868$. It can be seen that the F_F value on all metrics is greater than the critical value, so the original hypothesis can be rejected on all metrics, that is, the average rank of all algorithms is not the same on each metric. If the hypothesis that all comparing methods have equal performance is rejected, we conduct a post-hoc test, that is, Nemenyi test, to analyze the performance of all comparison algorithms. The formula is defined as [47]:

$$CD_\alpha = q_\alpha \sqrt{\frac{k(k+1)}{6N}}.$$

In this paper, $k = 9, N = 9$, thus $q_\alpha = 3.1017$. Therefore, $CD_a = 4.0043$. The results of different evaluation metrics are displayed in Fig. 3, which shows the relative performance of MLDRS and other comparing algorithms. The following conclusions can be obtained: in general, MLDRS algorithm outperforms other multi-label learning algorithm in seven evaluation metrics; for Hamming Loss, MLDRS performs the best statistically; in terms of Ranking Loss and Average Precision, MLDRS has the best performance against RF-ML and PPTReliefF; and MLDRS outperforms other algorithms in Ranking Loss, Coverage and Average Precision except the MVLD algorithm in statistical performance; for One Error, MLDRS is not statistically different from other algorithms except the PPTReliefF and RReliefF; however, in terms of Macro-F1 and Micro-F1, although MLDRS is ranked higher than some algorithms on average, there is still no evidence to show a statistically significant difference compared to all the other algorithms to show that MLDRS performs better than the others.

With regard to the whole metrics, compared with ML-KNN, RF-ML, PPTReliefF, MLNB, RRelief, MDFS, JFSC and MVLD, the proposed algorithm has achieved some excellent performance on nine datasets. The effectiveness and feasibility of the algorithm are further verified by experimental comparison analysis.

5.4. Experiments study on noisy label space

To further clarify the effect of MLDRS on the classification performance of the label importance distribution in a noisy label space, we tested whether MLDRS has a certain anti-noise ability.

We believe that we can add noise to the label space in two dimensions: sample and label dimensions. Adding noise based on the sample dimension involves selecting samples with different proportions and randomly adding noise to the label space of these samples. Adding noise based on label dimension involves selecting the number of labels in different proportions to add noise to these labels. To obtain more complex noise, we consider these two dimensions simultaneously and invert the labels as noise in the label space.

We randomly selected 10%, 30%, and 60% instances of the training set and added noise to its label space. We divide any 10% instances into two parts. In the first part, any label is selected to increase the noise, and in the second part, any two labels are selected to increase the noise. For any 30% instances, we divided them into three parts, one, two, and three labels were chosen to increase the noise in these pasts. Similarly, for any 60% instances, we divided them into five parts, successively increasing the number of optional labels to add noise.

In Fig. 4, we recorded the results of the ML-KNN and MLDRS algorithms for each evaluation metric for noise data under different label noise conditions for each data set. To facilitate image display, we further limit the coverage metric results to the range of [0, 1]. It can be observed that, as opposed to the superior performance of MLDRS in the Ranking Loss, One Error, and Coverage of Emotions dataset, the performance of MLDRS is suppressed and unsatisfactory after the addition of noise. This may be because the increased noise interferes with the selection of optimal features, making the final feature set suboptimal.

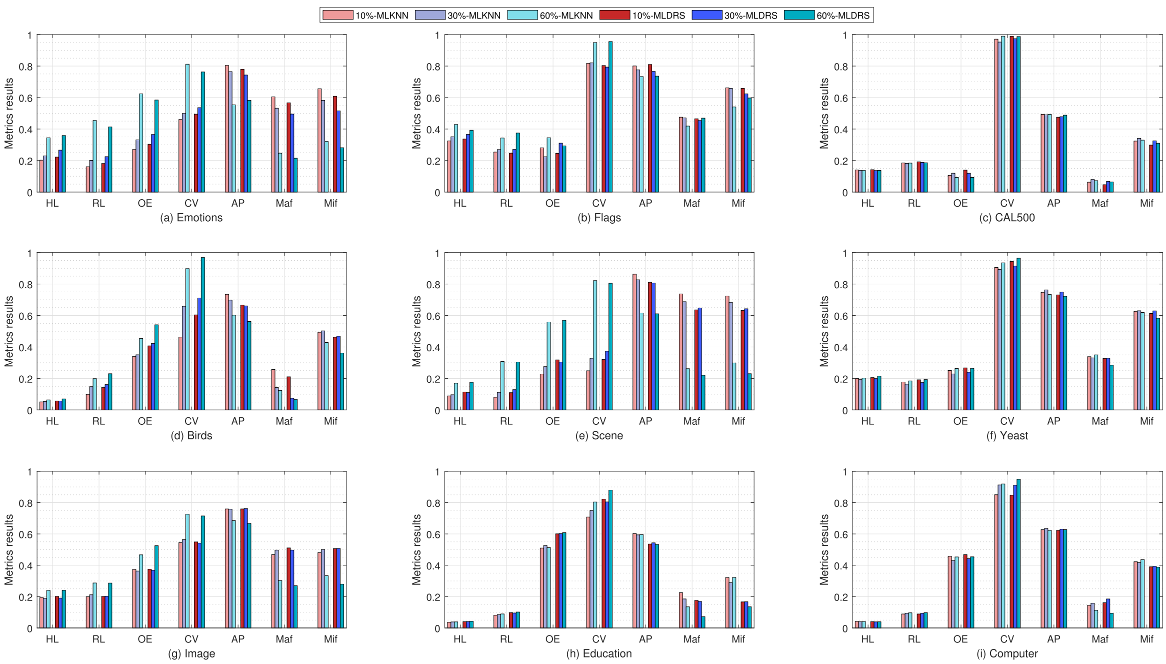

Fig. 4. Performance comparison in noise label space.

6. Conclusions

To handle multi-label classification tasks, we designed a novel feature selection method that considers the label significance distribution and inter-label correlations. In this study, we first granulated all instances and then calculated the importance distribution values of the labels of each instance in the label space. Consequently, MLDRS can explore more supervised information than traditional MLL methods. By improving the classical Apriori algorithm, we obtained a type of mixed-order inter-label relationship. Subsequently, a novel feature selection algorithm was designed based on combining inter-label correlations, which can obtain a feature subset from the raw feature space directly and retain the inherent meaning of the selected features. Finally, the experimental results show that the proposed algorithm is superior to the existing state-of-the-art MLL algorithms in terms of effectiveness and feasibility. In addition, we analyzed the performance of the algorithm in the case of increased noise in the label space. Although the algorithm does not show excellent performance in noisy conditions, this will be a question that we can consider further in future research.

In the future, we will continue to focus on the relationship between features and labels, and the relationships between labels. Inspired by feature fusion, futher research could investigate how to reduce the loss of feature information by fusing the most relevant features and establish a new feature space. Based on this feature space, we will further investigate MLL tasks.

CRediT authorship contribution statement

Yi Kou: Conceptualization, Methodology, Software. **Guoping Lin:** Data curation, Writing - original draft. **Yuhua Qian:** Formal analysis, Supervision. **Shujiao Liao:** Writing - review & editing.

Data availability

The authors do not have permission to share data.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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